

Ryan Peters

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SUMMARY

M.S. Computer Science student at Syracuse University focused on formal verification of distributed machine learning training systems, with a strong track record of building and shipping applied ML and AI applications. Experienced IT service analyst who supervises and trains student staff and improves support operations.

EDUCATION

Syracuse University, Syracuse, NY
M.S. in Computer Science, expected 2028
B.S. in Computer Science, [2025]

TECHNICAL SKILLS

Languages: Python, Racket and Scheme, C and C++, LLVM IR, TypeScript, JavaScript, OCaml (learning)
Machine Learning and AI: XGBoost, time-series modeling, retrieval-augmented generation, multi-agent systems, LLM fine-tuning (LoRA, Unsloth, Ollama), local LLM deployment (llama-cpp-python, GGUF), Sentence Transformers, spaCy, BLIP, EasyOCR
Formal Methods: TLA+, model checking, temporal logic, the mu-calculus, bisimulation, process algebra, operational semantics
Data and Tools: pandas, NumPy, ChromaDB, Streamlit, Docker, Git, GitHub Actions, Jira, Confluence, Astro, Jupyter

EXPERIENCE

Support Analyst II, Syracuse University ITS Service Center, Syracuse, NY
[Start year] to Present

- Run front-line and escalated help desk operations supporting students, faculty, and staff across the university.
- Supervise, train, and schedule student technology consultants, including coordinating multi-round hiring cycles.
- Manage Jira ticket queues and maintain support workflows and documentation in Confluence.
- Designed U-Build-It, an appointment-based technology coaching service for students.

SELECTED PROJECTS

Cross-Market Agricultural Commodity Prediction Engine (Python, XGBoost, Streamlit, llama-cpp-python)
Lead developer on an explainable-AI dashboard for commodity and livestock markets. Built a real-time XGBoost forecasting engine with automated feature engineering and time-series cross-validation, a cross-market correlation engine over aligned USDA, CME, and FRED data, and plain-language briefings from a privacy-preserving local quantized Llama-3-8B. Hardened the system with a 75-check automated validation suite.

Multi-Agent Restaurant and Activity Finder (Python, OpenAI, ChromaDB, Streamlit)
Built a chat-first, location-aware recommender on a multi-agent architecture, with an orchestrator coordinating parallel agents, a scrape-to-vector-store retrieval pipeline, and anti-hallucination prompting that constrains each agent to provided data.

AI Education Tool (Python, BLIP, EasyOCR)
Contributed memory and performance work to a multimodal tool that turns document pages into audio, including lazy loading of vision and OCR models to cut startup memory, a redesigned audio interface, and robust error handling.

Collaborative Media Review Tool (Vue 3, Pixi.js, Docker)
Built a SyncSketch-style frame-accurate media review and annotation app with WebGL canvas rendering and a containerized setup.